The documentation and process conversion measures necessary to comply with this revision shall be completed by 1 December 1989

INCH-POUND

MIL-S-19500/304C
6 June 1989
SUPERSEDING
MIL-S-19500/304B
14 September 1982

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
FAST-RECOVERY
TYPES 1N3885, 1N3886, 1N3888, 1N3890, 1N3891,
1N3893, 1N3890R, 1N3891R, 1N3893R AND A-VERSIONS
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for 12 and 20 ampere, silicon, semiconductor, fast recovery, power rectifier diodes. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figures 1 and 2 (DO-4).

1.3 Maximum ratings.

Type	V_R	V_{RWM}	I_O 1/ T_C = 100°C	I_{FSM} T_C = 100°C t = 1/120 s	t_{rr}	T_C
	V	V (pk)	A dc	a (pk)	ns	°C
1N3885	100	100	12	150	200	-65
1N3885A	100	100	20	225	150	
1N3890, R	100	100	12	175	200	
1N3890A, AR	100	100	20	250	150	
1N3886	200	200	12	150	200	
1N3886A	200	200	20	225	150	
1N3891, R	200	200	12	175	200	to
1N3891A, AR	200	200	20	250	150	
1N3888	400	400	12	150	200	
1N3888A	400	400	20	225	150	
1N3893, R	400	400	12	175	200	
1N3893A, AR	400	400	20	250	150	+150

1/ Derate linearly, 2 percent of I_O/°C for T_C > 100°C.
Storage temperature:
-65°C to +150°C - 1N3885, 1N3886, 1N3888, and A-versions
-65°C to +175°C - 1N3890, 1N3891, 1N3893 and A, R, AR versions
Barometric pressure reduced: 15 mmHg.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Electronics Research and Development Command, DRDEL-ED, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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1.4 Primary electrical characteristic.

Type	I_{R1}		I_{R2} T_A = 150 °C	
	VR (V dc)	µA dc	VR (V dc)	mA dc
1N3885, A, 1N3890, A, 1N3890R, AR	100	10.0	100	2
1N3886, A, 1N3891, A, 1N3891R, AR	200	10.0	200	2
1N3888, A, 1N3893, A, 1N3893R, AR	400	10.0	400	2

Type	$R_{\theta JC}$ °C/W
Isolated non - A	3.0
Isolated A	2.5
Non isolated non - A	2.0
Non isolated A	1.5

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATON

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification For.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

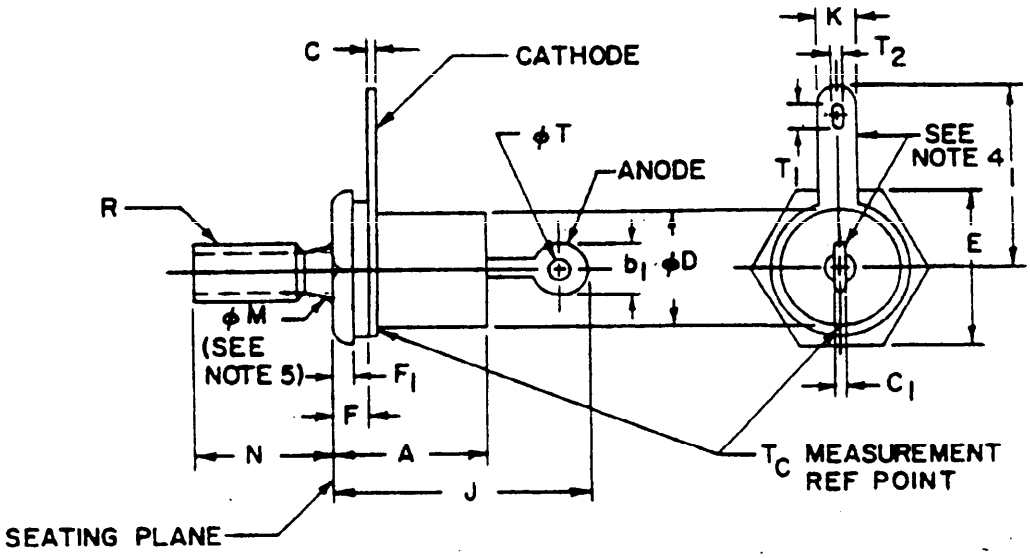
MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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Ltr	Dimensions		Notes
	Minimum	Maximum	
A		.550 (13.97)	
ϕD		.487 (12.37)	
E	.487 (12.37)	.500 (12.70)	
F_1	.040 (1.02)		
F	.085 (2.16)	.160 (4.06)	
b_1		.250 (6.35)	4
c	.020 (0.51)	.038 (0.97)	
I	.750 (19.05)	.875 (22.22)	

FIGURE 1. Physical dimensions of isolated types 1N3885, 1N3885A, 1N3886, 1N3886A, 1N3888, and 1N3888A.

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Ltr	Dimensions		Notes
	Minimum	Maximum	
J		.950 (24.13)	
K	.110 (2.79)	.140 (3.56)	4
T_2	.055 (1.40)	.075 (1.90)	
T_1	.110 (2.79)	.140 (3.56)	
N	.422 (10.72)	.453 (11.51)	
C_1		.050 (1.27)	
R			6,7,8,9
øT	.060 (1.52)		
øM	.163 (4.14)	.189 (4.80)	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Angular orientation of terminals is undefined. Square or radius on end of terminals is optional.
5. Diameter variations within these limits are permitted.
6. The ANSI thread reference is .190-32 UNF-2A.
7. Maximum pitch diameter of plated threads shall be basic pitch diameter .169 inch (4.29 mm) and in accordance with FED-STD-H28.
8. Unit shall not be damaged by torque of 15 inch-pounds applied to .190-32 UNF-2B nut assembled on thread.
9. Complete threads shall extend to within .078 inch (1.98 mm) of the seating plane.
10. For marking, see 3.4.
11. Stud and seating plane shall be electrically insulated from the case, cathode, and anode.

FIGURE 1. Physical dimensions of isolated types 1N3885, 1N3885A, 1N3886, 1N3886A, 1N3888, and 1N3888A - Continued.

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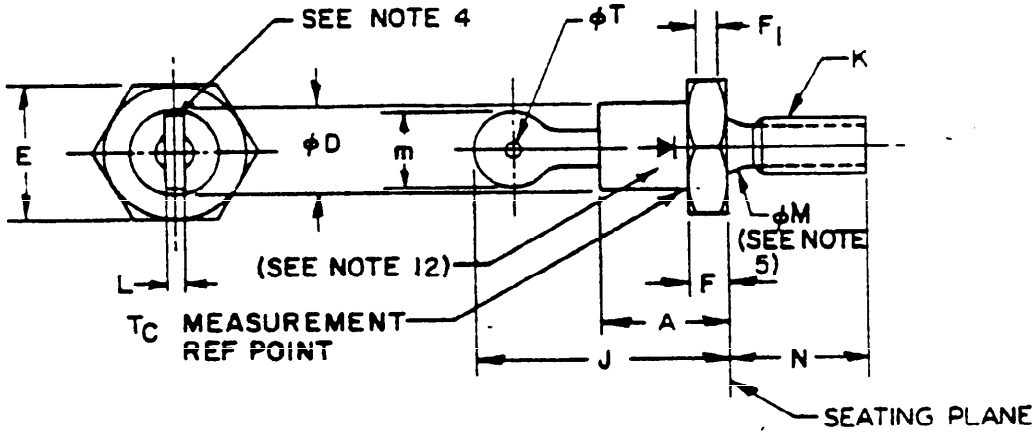


FIGURE 2. Physical dimensions of non-isolated types 1N3890, 1N3890A, 1N3891, 1N3891A, 1N3893, 1N3893A, 1N3890R, 1N3890AR, 1N3891R, 1N3891AR, 1N3893R, and 1N3893AR (DO-4).

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Ltr	Dimensions		Notes
	Minimum	Maximum	
A		.405 (10.29)	
øD		.424 (10.77)	
E	.424 (10.77)	.437 (11.10)	
F	.075 (1.90)	.175 (4.44)	
J		.800 (20.32)	
m		.250 (6.35)	10
F	.060 (1.52)		
øT	.060 (1.52)		
N	.422 (10.72)	.453 (11.51)	
K			6,7,8,9
L	.020 (0.51)	.065 (1.65)	
øM	.163 (4.14)	.189 (4.80)	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Angular orientation of terminals is undefined.
5. Diameter variations within these limits are permitted.
6. The ANSI thread reference is .190-32 UNF-2A.
7. Maximum pitch diameter of plated threads shall be basic pitch diameter .169 inch (4.29 mm) and in accordance with FED-STD-H28.
8. Unit shall not be damaged by torque of 15 inch-pounds applied to .190-32 UNF-2B nut assembled on thread.
9. Complete threads shall extend to within .078 inch (1.98 mm) of the seating plane.
10. Terminal-end shape is unrestricted.
11. Reversed (anode to stud) units shall be marked with an "R" following the last digit in the type number.
12. Forward polarity (cathode to stud) marking is shown.

FIGURE 2. Physical dimensions of non-isolated types 1N3890, 1N3890A, 1N3891, 1N3891A, 1N3893, 1N3893A, 1N3890R, 1N3890AR, 1N3891R, 1N3891AR, 1N3893R, and 1N3893AR (DO-4) - Continued.

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3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein. Lot accumulation period shall be three months in lieu of six weeks.

3.2 Abbreviations, symbols and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500, and as follows:

R_{ISO} - - - - - - - - - Insulation resistance case-to-stud.

$i_r(rec)$- - - - - - - - Peak reverse recovery current.

3.3 Design, construction and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figures 1 and 2 herein.

3.3.1 Diode types 1N3885, 1N3886, and 1N3888 (see figure 1). Diode types 1N3885, 1N3886, and 1N3888 have the stud and seating plane electrically insulated from the anode, cathode, and case.

3.3.2 Diode types 1N3890, 1N3891, and 1N3893 (see figure 2). Diode types 1N3890, 1N3891, and 1N3893 (forward polarity) have the cathode electrically connected to the stud and case.

3.3.3 Diode types 1N3890R, 1N3891R, and 1N3893R (see figure 2). Diode types 1N3890R, 1N3891R, and 1N3893R, (reverse polarity) have the anode electrically connected to the stud and case.

3.3.4 Dissimilar construction. Types utilizing construction as shown on figure 1 shall not be considered structurally identical to types utilizing construction as shown on figure 2.

3.4 Marking. Marking shall be in accordance with MIL-S-19500, except at the option of the manufacturer, the following marking may be omitted from the body of the device.

3.4.1 Polarity. The polarity shall be indicated by a graphic symbol with the arrow pointing toward the negative end for forward bias. The reversed units shall also be marked with an R following the last digit in the type number.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualificaton inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein. Tests in either polarity shall be sufficient to obtain qualification approval of both polarities.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II) and specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

4.3.1 Thermal response (ΔV_F measurements). The delta V_F measurements shall be performed in accordance with method 3101 of MIL-STD-750. The delta V_F conditions and maximum V_F limit shall be derived by each vendor. The chosen ΔV_F measurements and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen ΔV_F value shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. t_p = 250 ms heating current (I_H) $\geq$ rated I_O.

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4.3.2 Surge current. Surge current, see MIL-STD-750, method 4066. I_O = 0; $V_{RM}(W)$ = 0; T_A = 25°C; 6 surges. t_p = 8.3 ms, one surge per minute max.

Surge current value (I_{FSM})		
	A-version	Non-a version
1N3885	350	250
1N3886	350	250
1N3888	350	250
1N3890	500	400
1N3891	500	400
1N3893	500	400

Screen (see table II of MIL-S-19500)	Measurements	
	JANS level	JANTX and JANTXV levels
1/	Surge see 4.3.2 Thermal response see 4.3.1	Surge see 4.3.2 Thermal response see 4.3.1
4	N/A	N/A
9	V_{F2} and I_{R1}	V_{F2} and I_{R1}
10	MIL-STD-750, method 1038 test condition A; t = 96 hrs	MIL-STD-750, method 1038 test condition A; t = 48 hrs
11	V_{F2} and I_{R1}; ΔV_{F2} = +0.1 V (pk); ΔI_{R1} = 5 µA dc or 100% from the initial value whichever is greater. Subgroup 2 of table I herein.	Subgroup 2 of table I herein. 3/ V_{F2} and I_{R1}; ΔV_{F2} = +0.1 V (pk); ΔI_{R1} = 5 µA dc or 100% from the initial value whichever is greater.
12	Burn-in see 4.3.3 and 4.5.1 MIL-STD-750, method 1038 test condition B	Not applicable
13	Subgroups 2 and 3 of table I herein; ΔV_{F2} = +0.1 V (pk); ΔI_{R1} = 5 µA dc or 100% from the initial value whichever is greater.	Not applicable

1/ Surge shall precede thermal response. These tests shall be performed anytime after screen 3 and before screen 9.
2/ I_{R1} measurement shall not be indicative of an open condition.
3/ PDA of screen 13 shall apply to screen 11, for JANTX and JANTXV.

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4.3.3 Burn-in and life tests. Power burn-in conditions are as follows:

MIL-STD-750, method 1038, test condition B. T_C = 100°C, Rated I_O, see 4.5.1
1N3885, 1N3890, 1N3890R - - - - V_R = 100 V (pk)
1N3886, 1N3891, 1N3891R - - - - V_R = 200 V (pk)
1N3888, 1N3893, 1N3893R - - - - V_R = 400 V (pk)

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table V herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and table IIa and table IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows:

4.5.1 Burn-in and steady-state operation life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectifier current. The forward conduction angle of the rectified current shall not be greater than 180°C nor less than 150°C.

4.5.2 Insulation resistance (case-to-stud). The case-to-stud insulation resistance shall be determined in terms of current flow between the case and stud. The specified test method shall be utilized except that the applied voltage shall be within ±2.0 percent of the specified value and the current shall be measured by a series microammeter with a voltage drop at 1.0 µA dc of less than 1.0 V dc, and an accuracy of ±2.0 percent at the 1.0 µA dc point.

4.5.3 Reverse recovery time. The reverse recovery time shall be measured in the circuit of figure 4 or equivalent. Care should be exercised to minimize stray inductances in the test circuit and to insure that the total resistance of the reverse current loop can be adjusted sufficiently low so that more than 2 amperes will flow if not blocked by the diode being tested. Switch SW shall be activated and the regulated voltage source adjusted to achieve the specified forward current when SW is open. Inductor L and resistor R shall be adjusted to achieve the following characteristics of the waveform.

a. The di/dt shall be the specified value between the forward 0.5 ampere point and the reverse 0.2 ampere point.
b. The $i_{r(rec)}$ shall be the maximum value obtainable, except that if it exceeds 2 amperes, it shall be reduced to equal 2 amperes.

The reverse recovery time shall then be determined from the current waveform as shown on figure 4.

4.5.4 DC intermittent operation life. A cycle shall consist of an "on" period, when power is applied suddenly, not gradually, to the device for the time necessary to achieve a delta case temperature of 85°C +15°C -5°C, followed by an "off" period, when the power is suddenly removed, for cooling. Auxiliary (forced) cooling is permitted during the off period only.

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30 s $\leq t_{heating} \leq$ 60 s. $P = V_F \times I_F$ or $P = V_{F(pk)} \times I_{av}$ if using sine wave current.

DC full wave current shall be used for the power required during the "on" period (or equivalent half sine wave current).

Within the time interval of 50 cycles before to 500 cycles after the termination of the test, the sample units shall be removed from the specified test conditions and allowed to reach room ambient test conditions. Specified end-point measurements for qualification and quality conformance inspection shall be completed within 96 hours after removal of sample units from the specified test conditions. Additional readings may be taken at the discretion of the manufacturer.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD		Symbol	Limits		Units
	Method	Conditions	1/ JANS	JAN 1/ JANTX JANTXV		Min	Max	
Subgroup 1								
Visual and mechanical inspection	2071							
Subgroup 2								
Forward voltage	4011	I_f = 38 a (pk) $t_p \leq$ 8.3 ms, duty cycle $\leq$ 2% maximum			V_{F2}		1.50	V (pk)
Forward voltage	4011	$I_F = I_{FSM}$ t_p = 800 µs			V_{F1}		2.75	V dc
Reverse current	4016	DC method V_R = method V_R = (dc) (see 1.3)			I_{R1}		10	µA dc
Insulation resistance (case to stud) applicable only to 1N3885, 1N3885A 1N3886, 1N3886A 1N3888, 1N3888A	1016	Test condition C See 4.5.2			R_{ISO}	10^9		ohms
Subgroup 3								
High temperature reverse current operaion	4016	$T_A = T_C$ = 150°C DC method V_R = rated V_R (see 1.3)			I_{R2}		2	mA dc
Subgroups 4 and 5								
Not applicable								
Subgroup 6								
Not applicable								
Subgroup 7								
Reverse recovery time	4031	T_C = 55°C I_F = 1 A dc, V_R = 30 V dc di/dt = -25 A/µs $i_{r(rec)} \leq$2 a (pk) (See 4.5.3)			t_{rr}			ns
Non A							200	
A-versions							150	

1/ For LTPD numbers see MIL-S-19500.

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TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figures 1 and 2.		
Subgroup 2 2/				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051			
Surge	4066	T_C = 100°C; t_p = 8.3 ms (see 1.3) V_R = rated V_R six 1/120 s. surges; 1 surge/minute max		
Non-A version devices		i_f (surge) = 150 A dc I_O = 12 A dc		
A version devices		i_f (surge) = 225 A dc I_O = 20 A dc		
Hermetic seal	1071			
Fine leak Gross leak				
Electrical measurements		See table V, steps 1, 2, 3 and 4		
Decap-internal visual (design verification)	2075			
Bond strength (wire or clip bonded devices only)	2037			
Die shear	2017			
Subgroup 4			5	12/0
DC intermittent operation life	1037	.25 rated $I_O \leq I_O$ applied $<$ rated I_O (see 4.5.4) 2,000 cycles		
Electrical measurements		See table V, steps 1, 2 3, 4		
- - -				

See footnotes at end of table.

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sure. Contact with galvanized iron, and copper and its ... silver positive pres

TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 5				
Accelerated steady-state operation life	1027	I_F = 35 A dc at; T_A = 125°C for 96 hrs, or adjusted as required by the chosen T_A to give an average lot T_J = 275°C; f = 60 Hz V_R = Rated Vr (pk) (see 1.3)		
Electrical measurements		See table V, steps 1, 5 and 6		
Bond strength (Al-Au die interconnect only)	2037			
Subgroup 6				
Thermal resistance	3101	See 1.4 See figure 3		

1/ For LTPD numbers see MIL-S-19500

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TABLE IIb. Group B inspection for JAN, JANTX, and JANTXV.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1 2/				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 2				
Thermal shock (temperature cycling)	1051			
Insulated types		-65°C to +150°C		
Uninsulated types		-65°C to +175°C		
Surge	4066	T_C = 100°C; V_R = Rated V_R (see 1.3); six surges; 1 surge/minute max t_p = 8.3 ms		
Non-A version devices		i_f (surge) = 150 A dc I_O = 12 A dc		
A version devices		i_f (surge) = 225 A dc I_O = 20 A dc		
Hermetic seal	1071			
Fine leak Gross leak				
Electrical measurements		See table IV, steps 1, and 2		
Subgroup 3 2/				
DC intermittent operation life	1037	.25 rated $I_O \leq I_O$ applied < rated I_O (see 4.5.4) 2,000 cycles		
Electrical measurements		See table V, steps 1, 2 and 3		
Steady-state dc blocking life	1038 or 1049	Condition A, 340 hrs		
Electrical measurements		See table V, steps 1, and 2		
- - -				

See footnotes at end of table.

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TABLE IIb. Group B inspection for JAN, JANTX, and JANTXV - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 4				
Decap internal visual (design verification)	2075			
Subgroup 5				
Not applicable				
Subgroup 6				
High-temperature (nonoperating)	1032	T_A = 150°C		
Electrical measurements		See table V, steps 1, and 2		

1/ For LTPD numbers see MIL-S-19500.

2/ A separate sample may be pulled for each test.

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TABLE III. Group C inspection (all quality levels).

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figures 1 and 2		
Subgroup 2				
Thermal shock (glass strain)	1056	Condition B		
Terminal strength (tension)	2036	Test condition A; t = 15 s		
1N3885, A 1N3886, A 1N3888, A		Weight = 5 lbs.		
1N3890, A, R, AR 1N3891, A, R, AR 1N3893, A, R, AR		Weight = 20 lbs.		
Bending stress		Test condition F method B; t = 15 s.		
1N3885, A 1N3886, A 1N3888, A		Weight = 1 lbs.		
1N3890, A, R, AR 1N3891, A, R, AR 1N3893, A, R, AR		Weight = 5 lbs.		
Seal torque		Test condition D1; Torque = 10 oz-in, t = 15 s.		
Hermetic seal	1071			
Fine leak Gross leak				
Moisture resistance	1021			
Electrical measurements		See table V, steps 1, 2 and 3 (JAN JANTX and JANTXV) steps 1, 2, and 3 (JANS)		
- - -				

See footnote at end of table.

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TABLE III. Group C inspection (all quality levels) - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 3				
Shock	2016			
Vibration, variable frequency	2056			
Constant acceleration	2006			
Electrical measurements		See table V, steps 1, and 2		
Subgroup 4				
Salt atmosphere (corrosion)	1041			
Subgroup 5				
N/A				
Subgroup 6			5	12/0
DC intermittent operation life	1037	.25 rated $I_o \leq I_o$ applied $\leq$ rated I_o (see 4.5.4) 6,000 cycles		
Electrical measurements		See table V, steps 1, 3, 5 (JANS), and 6 and steps 1, 3 and 6 (JAN, JANTX, and JANTXV)		

1/ For LTPD numbers see MIL-S-19500.

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TABLE IV. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD	Small lot quality conformance inspection n/c
	Method	Conditions		
Subgroup 1			10	8/0
Thermal shock (glass strain)	1056	0°C to 100°C, 100 cycles		
Hermetic seal	1071			
Electrical measurements		See table V, steps 1, 2, and 3		
Subgroup 2			5	12/0
Steady-state dc blocking life	1038	Condition A; 1,000 hrs.		
Electrical measurements		See table V, steps 1, 5 and 6		
Subgroup 3			3/0	1/0
DPA		Photos of cross sections shall be submitted in the qualification report. Vendors shall retain duplicate photos.		
Subgroup 4			10	8/0
Thermal resistance	3101	See 1.4 and figure 3		
Subgroup 5				
Barometric pressure (reduced)	1001	15 mm Hg ≤ Pressure; t = 60 s. While the test is being performed I_R shall be monitored and shall not exceed 25 µA dc		

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TABLE V. Groups A, B, and C electrical end-point measurements.

Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1.	Forward voltage	4011	I_F = 38 a (pk) (pulse); pulse width = 8.3 ms duty cycle = 2% maximum	V_{F2}		1.50	V (pk)
2.	Reverse current	4016	DC method V_R = Rated V_R (dc) (see 1.3)	I_{R1}		10	µA dc
3.	Thermal response	3101	See 4.3.1	ΔV_F			mV dc
4.	Forward voltage	4011	I_F = 50 mA dc	ΔV_{F3}	+50 mV dc maximum change from previous to post intermittent life and thermal shock measurement tests (JANS, only)		
5.	Reverse current	4016	DC method V_R = Rated V_R (dc) (see 1.3)	ΔI_{R1} 1/	100% or ±5 µA, (whichever is greater) change from initial group A reading		
6.	Reverse current	4016	DC method V_R = Rated V_R (dc) (see 1.3)	I_{R3}		20	µA dc

1/ Devices which exceed the group A limits from this test shall be rejected.

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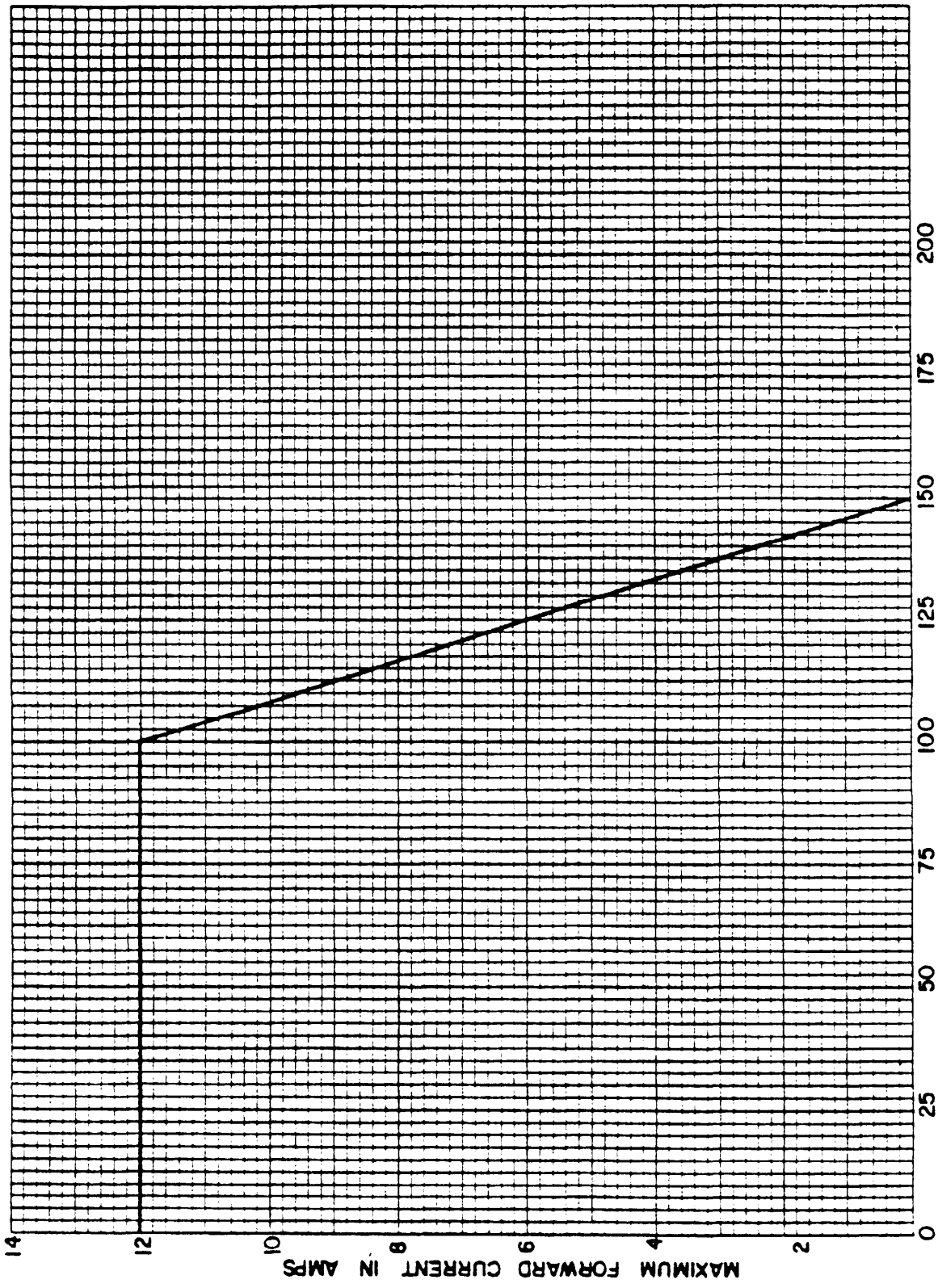
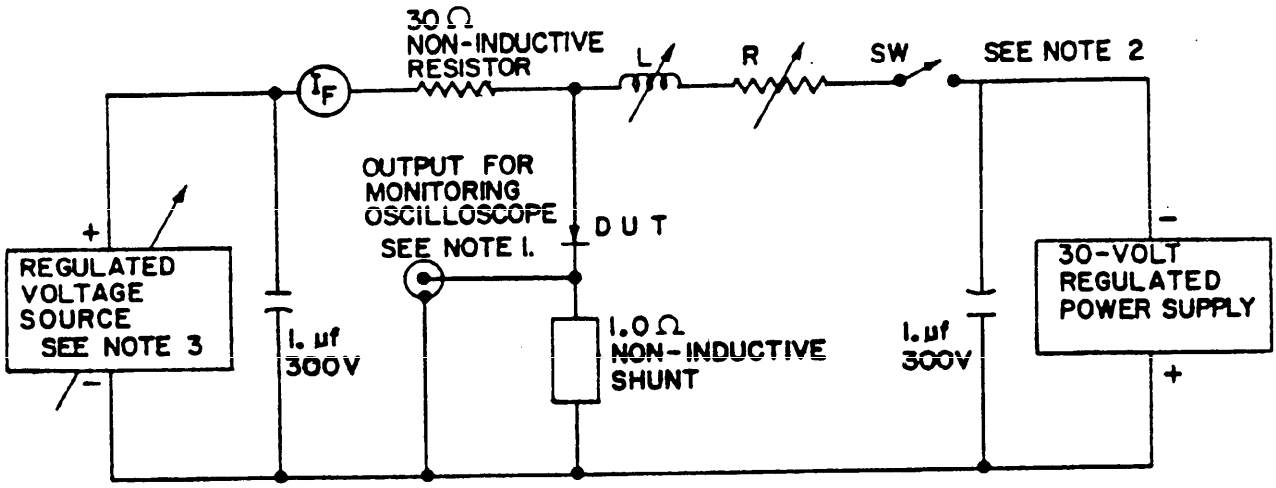


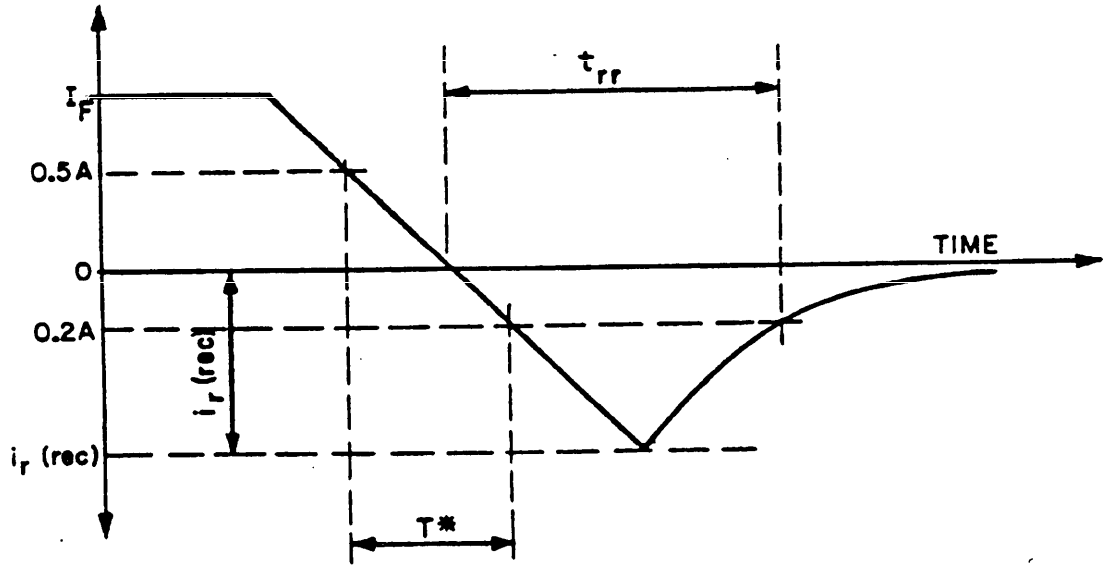
FIGURE 3. Maximum case temperature in °C.

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NOTES:
1. Monitoring oscilloscope requirements: $t_r \leq 14$ ns, $R_{in} \geq 9$ MΩ, $C_{in} \leq 12$ pF, L_{in} (series) ≤ 0.5 µH.
2. SW characteristics: Mercury-wetted make-before-break relay switched at a 60 Hz rate. The relay should conduct for approximately 640 µs and be open for approximately 7.7 ms. (C.P. Clare HGP 1004 or equivalent).
3. Voltage source characteristics: Output impedance $\leq 0.5\Omega$ from 0 to 2 kHz.

FIGURE 4. Reverse recovery test circuit.



*Adjust L and R in accordance with paragraph 4.5.3 to achieve T = .028 µs (L=1.2 µH). Then, $\frac{di}{dt} = \frac{-0.7}{.028} = -25A/\mu s$.

FIGURE 5. Reverse recovery current waveform through device under test.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.3 Supersession data. The following types were superseded through a previous revision of this specification and the cancellation of MIL-S-19500/266. A-version devices may be substituted for Non A-version devices. Applicable substitutes are as follows:

Superseded types	Substitute types
1N3874, 1N3875, 1N3884, 1N3876	1N3885
	1N3886
1N3877, 1N3878, 1N3887	1N3888
1N3879, 1N3880, 1N3889	1N3890
1N3879R, 1N3880R, 1N3889R	1N3890R
1N3881	1N3891
1N3881R	1N3891R
1N3882, 1N3883, 1N3892	1N3893
1N3882R, 1N3883R, 1N3892R	1N3893R

NOTE: Superseded types shall not be used for new design.

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's part numbers. This information in no way implies that manufacturer's part numbers are suitable as a substitute for the military part number.

Supersession and cross reference data.

Military part number	Manufacturer's code CAGE	Manufacturer's and user's part number
1N3893		SR7001H

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 19, 85, 99
DLA - ES

Users activities:
Army - SM
Navy - AS, CG, MC, OS

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-1062)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-19500/304C	SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST RECOVERY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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